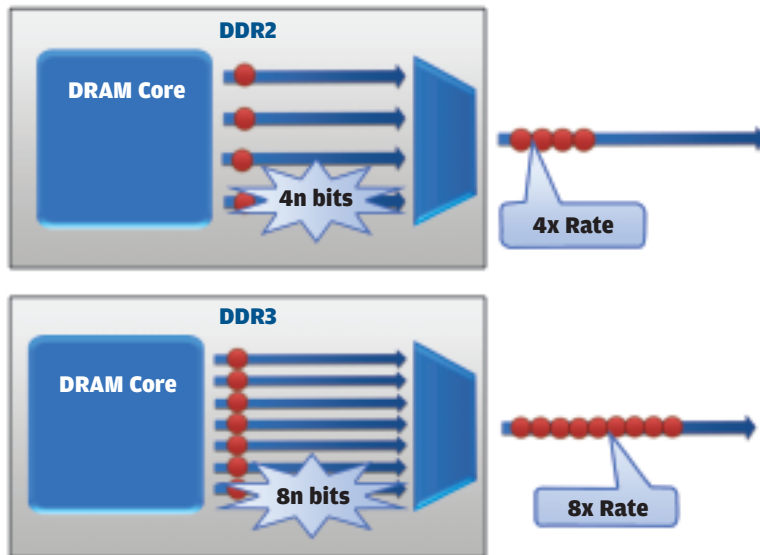


we saw memory modules with up to 2 GB of memory. Such modules use high density chips of capacities of up to 128 and even 256 MB (each memory module has either eight or sixteen such chips). With DDR2 manufacturers promise as



much as 512 MB on a single chip, meaning a module with a single bank of chips (chips on a single side of the module) could have a capacity of as much as 4 GB! We've heard of manufacturers working on chips with capacities of 1024 MB. Although double bank chips (16 GB on a single module!) will be prohibitively expensive, and will be a rarity, the prospect of seeing eight gigabytes on a single module is exciting. This would also drive prices lower. 2009 should see a sharp decline in DDR3 prices by our estimates—much like DDR2 saw back in mid 2007. Faster, low latency modules should also become available at the price of slower modules, and the ones already in the market will get cheaper. 2009 should be the time that users should migrate to DDR3, as of now DDR2 hasn't relinquished the throne in terms of market popularity.

In terms of performance things are about even, with fast, low latency DDR2 modules competing hard with similar DDR3 modules. Once 1.6 GHz and 1.8 GHz DDR3 modules become *de facto* however, DDR2 will trail behind.

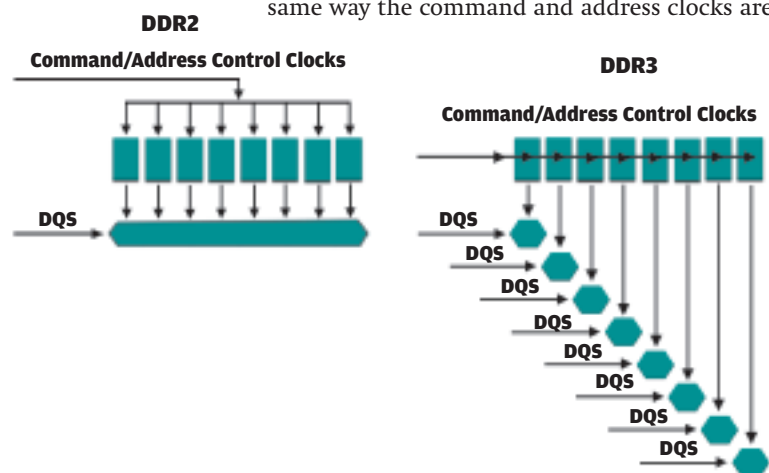
GDDR5—Speed Demon

After a hiatus of nearly 4 years, the world of video memory is about to receive its much awaited shot in the arm courtesy GDDR5. The current norm, GDDR3, has done a great job. It's gone from a mere 256 bits and 1 GHz in the GeForce 6800GT back in 2004, to a 512-bit 2.2 GHz part in the GeForce GTX280. However this industry is all about change, and we're seeing a transition of sorts about to commence. We've already seen GDDR5 make an entry in ATI's

latest Radeon HD4870. The memory on this card is built on a simple 256-bit bus. As opposed to NVIDIA's GTX280 which has a massive 512 bit memory interface. The aim of both vendors is simple—more bandwidth. However, the approach is radically different—NVIDIA uses wider memory buses, ATI has gone with GDDR5 which has higher clocks.

ATI's move makes sense because increasing the bus width is no easy or cost effective task. Each memory chip has to have more pins—this leads to a larger chip—which leads to fewer such chips per wafer (yes, memory chips are also cut from silicon wafers). This drives up the cost, which is why mainstream and budget graphics cards usually have 128- or 256-bit memory interfaces. Even bumping up clock speeds doesn't help, as thermal thresholds are eventually hit. GDDR5 is all about transmitting more data per pin, (we're talking about the pins on a BGA memory socket), per clock. GDDR5 has the benefit of requiring less power, and also fewer pins which simplifies the manufacturing process.

So what does GDDR5 do radically different from GDDR3 and GDDR4? First up, there's an exciting new feature called Clock Data Recovery (CDR hereon) which is all about maintaining clean signals while transporting data and thus preserving data integrity. With CDR the data stream itself generates the memory clock speed, and not the memory controller. The bus between the memory chips and the graphics processor is actually trained or synced. In the same way the command and address clocks are



also synchronised. For this reason this feature is also sometimes called Interface Timing, or Clock Training. The clock speed is actually picked up and read from the data stream and data and addresses and commands are adjusted or de-skewed (the technical term for it) for best possible data integrity. This is one of the new features integrated into GDDR5 memory modules to achieve higher clock speeds.

Traditionally GDDR3 and other memory technologies have used a single write clock (designated as WC) which transmits data and